

Qn 7	Answer	Marks
(a)(i)	At level flight, there is no resultant vertical force. Hence, lift = weight of aircraft = 1.5×10^6 N At constant velocity, the resultant force horizontally is zero. Hence, drag = thrust = 0.60×10^6 N	A1 A1
(a)(ii)	Taking moments about the intersection between the weight and thrust, Torque generated by the lift-weight couple = Torque generated by the thrust-drag couple $\Rightarrow 1.5 \times 10^6 \times 0.75 = 0.60 \times 10^6 \times d$ $\Rightarrow$ vertical separation $d = 1.9$ m	M1 A1
(a)(iii)	The <u>horizontal stabiliser</u> with its <u>vertical downward force</u> acting on it and (its long distance from the CG), would provide a (<u>clockwise</u>) <u>moment</u> about the CG to enable the plane to rotate against the downward pitch.	B1 B1
(b)(i)	Air gains momentum when it passes through the jet engine and hence by Newton's 2nd law, there is a force acting on the air by the jet engine. By Newton's 3rd Law, there is an equal in magnitude and opposite in direction force on the jet engine to move the jet engine forward. That is the thrust.	B1 B1
(b)(ii)	Force on jet engine by air = Force on air by the jet engine = rate of change of momentum of the air = $210 \text{ kg s}^{-1} \times 580 \text{ m s}^{-1} = 1.22 \times 10^5 \text{ N}$	M1 A1
(b)(iii)	Possible reasons include. - Combusted fuel is mixed with the gas before ejection and hence the actual momentum gain is larger. - Pressure difference between the air in front of engine and back of engine can contribute to an additional thrust - The engine needs to exert a forward force on the air to reduce the speed of the air before it is combusted and pressurized, the air exerts a backward force in return and this reduces the thrust. Do not accept : That there is some initial speed of the air as the question in (b)(ii) gives the speed gained.	B1

(c)	Each wing has to provide sufficient lift to support half of the weight. $(P_L - P_U)A = \frac{mg}{2}$ $\Rightarrow P_U = P_L - \frac{mg}{2A} = 7.00 \times 10^4 \text{ Pa} - \frac{(2.85 \times 10^5)(9.81)}{2(360)}$ $= 6.61 \times 10^4 \text{ Pa}$ [Only deduct one mark if mg is used instead.]	M1 A1												
(d)(i)	From the Lift Equation: $L = \frac{1}{2} C_L \rho v^2 S \Rightarrow C_L = \frac{2L}{\rho v^2 S}$ $\text{Units of } C_L = \frac{(\text{kg m s}^{-2})}{(\text{kg m}^{-3})(\text{m}^2 \text{ s}^{-2})(\text{m}^2)}$ $= 1$ Hence, C_L is a dimensionless quantity.	M1 A1												
(d)(ii)	The aircraft tilts so that the horizontal component of the lift force can provide for the (centripetal) force needed to make the turn. It needs to increase in speed for a larger the lift force ($L = \frac{1}{2} C_L \rho v^2 S$) so that the vertical component of the lift force can support the weight.	B1 B1												
(e)(i)	$\frac{L}{D} = \frac{\frac{1}{2} C_L \rho v^2 S}{\frac{1}{2} C_D \rho v^2} = \frac{C_L}{C_D}$	M1												
(e)(ii)	At maximum lift to drag ratio, most lift is generated (to support the weight) at minimum drag (<u>or</u> per unit drag). Hence, less fuel is wasted to overcome the drag force.	B1												
(e)(iii) 1.	For angle of attack $\alpha = 4^\circ$, $C_L = 0.45$ and $C_D = 0.0275$ $L/D = C_L/C_D = 0.45 / 0.0275 = 16.36$ Other possible combinations: <table border="1" data-bbox="228 1473 1117 1731"> <thead> <tr> <th>C_L</th><th>C_D</th><th>L/D</th></tr> </thead> <tbody> <tr> <td>0.45</td><td>0.03</td><td>15.00</td></tr> <tr> <td>0.425</td><td>0.0275</td><td>15.45</td></tr> <tr> <td>0.425</td><td>0.03</td><td>14.20</td></tr> </tbody> </table>	C_L	C_D	L/D	0.45	0.03	15.00	0.425	0.0275	15.45	0.425	0.03	14.20	M1
C_L	C_D	L/D												
0.45	0.03	15.00												
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(e)(ii)2.	Correct plot B1 A smooth best fit line through the plots with maximum at $\alpha = 4^\circ$ B1	B2												
	[Total : 22 marks]													